Fiscal Year:	FY 2024	**Task Last Updated:**	FY 03/10/2025
PI Name:	Hyers, Robert Ph.D.		
Project Title:	Non-Contact Measurement of Thermal Conductivity of Undercooled and Reactive Liquid Metals		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	01609-2280	**Congressional District:**	2
Comments:	Formerly with University of Massachusetts at Amherst.		
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2017/2019 Physical Sciences NNH17ZTT001N-17PSI-F: Use of the NASA Physical Sciences Informatics System – Appendix F
Start Date:	12/01/2021	**End Date:**	11/30/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC22K0207		
Performance Goal No.:			
Performance Goal Text:			
	We propose to apply new methods, tools, and models to the analysis of experiments from Microgravity Science Laboratory-1 (MSL-1) TEMPUS (Tiegelfreies Elektromagnetisches Prozessieren Unter Schwerelosigkeit; Electromagnetic Containerless Processing in Microgravity), which are in the NASA Physical Sciences Informatics (PSI) System. MSL-1 TEMPUS was a containerless processing facility which collected data on thermophysical properties of dozens of different metals and alloys on two Space Shuttle flights in 1997, STS-83 and STS-97. These experiments were designed to allow measurements that could not be made on Earth, and to set benchmarks for accuracy and range of measurements. The data from these experiments are still being used today in ongoing research. In addition, follow-on investigations currently flying on the International Space Station (ISS) have developed methods, tools, and numerical models that are significantly advanced from the state of the art in 1997.		

Task Description:	Experiments to measure the thermal conductivity of undercooled molten metals were performed as a part of this investigation, using modulation calorimetry. However, these experiments reported errors of greater than 100% vs. conventional measurements. These errors were attributed to magnetohydrodynamic (MHD) fluid flow in the sample. Our group has prepared a model of the MHD flow. Preliminary calculations on one of the experiments in PSI shows that we can recover the thermal conductivity with a precision more than an order of magnitude greater than previously reported. We propose to verify and validate the new tools on the MSL-1 data, and establish compatibility. The second objective, perhaps the more important, will unlock thermophysical property data gathered on NASA missions, which remains unpublished.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project will expand the capabilities of non-contact modulation calorimetry to measure the thermal conductivity liquid metals. This property is essential to accurately understanding and modeling the thermofluid response of liquid metals, including in casting, welding, and additive manufacturing.
Task Progress:	• The start of this project was significantly delayed by COVID. The proposed start date was 9/1/2020, but the start date actually awarded was 12/1/2021. • The proposed model was built in ANSYS Fluent. • The model was validated against the prior measurements of Wunderlich, et al., Appl. Phys. Lett. 62(24) 1993. • Results were presented at two large, international conferences. [Ed. Note: See Bibliography.] 1. G. Bracker and R.W. Hyers, "Convection during Modulation Calorimetry Experiments in Electromagnetic Levitation", presented at TMS 2023, 152nd Annual Meeting and Exhibition, San Diego, CA, March 23, 2023. 2. G. Bracker and R. Hyers, "Internal Heat Transfer and Convection during Modulation Calorimetry Experiments in Microgravity Electromagnetic Levitation", presented at FEMS EUROMAT 23: Materials Science and Technology in Europe, Frankfurt am Main, Germany, September 5, 2023. • The PI changed universities on January 1, 2023 to the Worcester Polytechnic Institute (WPI). The disruptions caused by this transition affected the timeline for reporting. • The model was successfully transferred to Ms. Lydia Ellen Tonani, PhD student at WPI, for application to new modulation calorimetry experiments on the International Space Station Electromagnetic Levitator (ISS-EML). • Additional work has continued at WPI under institutional support. • Requesting less than the standard 2-year period of performance, as we proposed, is not recommended.
Bibliography Type:	Description: (Last Updated: 03/30/2026)
Abstracts for Journals and Proceedings	Bracker G, Hyers RW. "Convection during modulation calorimetry experiments in electromagnetic levitation." TMS 2023. 152nd Annual Meeting and Exhibition. The Minerals, Metals & Materials Society, San Diego, CA, March 19-23, 2023. Abstracts. TMS 2023. 152nd Annual Meeting and Exhibition. The Minerals, Metals & Materials Society, San Diego, CA, March 19-23, 2023. , Mar-2023
Abstracts for Journals and Proceedings	Bracker G, Hyers R. "Internal heat transfer and convection during modulation calorimetry experiments in microgravity electromagnetic levitation." FEMS EUROMAT 23: Materials Science and Technology in Europe, Frankfurt am Main, Germany, September 3-7, 2023. Abstracts. FEMS EUROMAT 23: Materials Science and Technology in Europe, Frankfurt am Main, Germany, September 3-7, 2023. , Sep-2023